Comparing tail forecasts: the number of exceedances is the sample size

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Abstract

When two forecasts of a tail, such as prediction intervals at high coverage, extreme quantiles or expected shortfall (ES), are compared by a consistent score, the score difference is known in advance except on the days on which the outcome falls beyond a forecast. The number of these exceedances, not the number of observations, is the effective sample size. For intervals at miscoverage level α, the number of observations a comparison needs grows like $1/\alpha$ while the number of exceedances it needs does not, and for some pairs of forecasts no test can do better. A confidence sequence that is valid at all times and needs no distributional assumption attains this rate; Hoeffding-type procedures, calibrated by the range of the score difference, need $1/\alpha$ times more. When exceedances are scarce, the Diebold–Mariano test declares the less conservative forecaster better with at least, asymptotically, the probability of observing no exceedance, whichever forecaster is actually better; the usual small-sample corrections do not remove this, while a bootstrap-t version largely does. For (value-at-risk, ES) forecasts, a less conservative ES can be shown without assumptions to be worse on the days on which it is less conservative, but never to be better without an assumption on the loss tail. In 250-day windows of 97.5% forecasts for seven stock indices, 18.4% of significant Diebold–Mariano results favouring the less conservative model contradict the ranking on the remaining days, against 6.9% of those favouring the more conservative one.

Keywords: expected shortfall; interval score; quantile score; forecast comparison; Diebold–Mariano test; confidence sequences; comparative backtesting.

1 Introduction

Forecasts of rare events are issued as prediction intervals with high nominal coverage, as quantiles at extreme levels, or as expected shortfall (ES): 99% intervals for electricity load or prices, the outer quantiles of the weighted interval score used in epidemic forecasting [7], and value-at-risk (VaR) and ES at the 97.5% level, on which market-risk capital rests [4]. Competing forecasters are compared through consistent scores: the interval score [22, 40],

$$\mathrm{IS}_\alpha(l, u; y) = (u - l) + \frac{2}{\alpha}(l - y)_+ + \frac{2}{\alpha}(y - u)_+, \tag{1}$$

the quantile (pinball) score $\rho_\tau(y - q)$ with $\rho_\tau(x) = x\{\tau - \mathbf{1}(x < 0)\}$, and, for ES jointly with VaR, the scores of Fissler and Ziegel [15] used in comparative backtests [33]. The usual tool is a test of equal predictive ability on the score differences [12, 20], and practitioners know that tail scores are noisy: tail-quantile scores "suffer from high variance due to the sparsity of observations" [8].

This paper makes that observation precise. When the outcome falls inside two intervals, their score difference equals the difference of their widths, which is known before the outcome; it departs from that value only on rounds in which an interval is missed (Lemma 1). The same holds for quantiles and for (VaR, ES) pairs with exceedances in place of misses. Comparisons therefore rest on the exceedances. A 250-day backtesting window at the 97.5% level contains about six if the VaR forecast is calibrated; a comparison of 99% intervals over ten thousand rounds rests on little more than a hundred. We call a procedure *anytime-valid* if its error guarantee holds simultaneously at all sample sizes, so that the comparison can be monitored after every round and stopped at any time.

We make three contributions.

1. *How much data a comparison needs* (Sections 3 and 4). For some pairs of interval forecasts, every test needs a number of misses that does not shrink with α, and hence a number of rounds that grows like $1/\alpha$, even for bounded outcomes. An empirical-Bernstein confidence sequence, valid at all times with no assumption on the outcomes, attains this rate up to constants and a logarithmic factor. Procedures calibrated by the range of the score difference need a factor of order $1/\alpha$ more.

2. *Why the Diebold–Mariano test misleads* (Section 5). When misses are scarce it declares the narrower, or less conservative, forecaster significantly better, whichever forecaster is better, with probability at least, asymptotically, that of observing no miss; no test that controls its error can reject this often.

3. *What can be certified for ES* (Section 6). The score difference is unbounded on one side whenever two ES forecasts differ. A less conservative ES forecaster can be shown to be worse without assumptions; no test can show it to be better without an assumption on the loss tail.

Simulations (Section 7) confirm the rates and show the Diebold–Mariano test, its usual small-sample corrections and an asymptotic confidence sequence losing error control at small α, while a bootstrap-t version of the test largely keeps it. Two applications follow: six online conformal methods on seven forecasting data sets at levels from 80% to 99% (Section 8), and six (VaR, ES) models on seven stock indices (Section 9), where Diebold–Mariano tests in 250-day windows mostly favour the less conservative model and then often contradict the ranking on the remaining days. Results stated for intervals hold for quantile scores at tail level τ with α replaced by 2τ, up to constants (Corollary 1); proofs are in the Appendix.

Related work. Tests of equal predictive ability are reviewed by Diebold [11]; small-sample versions use the correction of Harvey et al. [25] or fixed-smoothing asymptotics [10], and the model confidence set [24] extends the comparison to many forecasters. All rely on a central limit theorem for the mean score difference, which Section 5 shows to fail when exceedances are scarce. Rankings by consistent scores can depend on the score chosen, which Murphy diagrams display [13], and evaluation conditioned on extreme outcomes is misleading [30]; we fix the score and condition on nothing. For ES, Fissler et al. [16] and Nolde and Ziegel [33] recommend comparative backtests and note that rankings at extreme levels need long samples. Bauer [5] found by simulation that Diebold–Mariano tests of VaR and ES forecasts at the 1% level have low power and wrong-sign rejections favouring the less conservative forecast; Proposition 2 gives a mechanism for the second finding. Frederiksen et al. [17] study a different failure, score differences with infinite variance. Jiao et al. [28] construct comparative e-backtests for risk measures including ES under a known bound on losses; Proposition 5 shows that some such assumption is unavoidable in one direction. Confidence sequences for means go back to the nonparametric constructions of Howard et al.

[27] and the betting approach of Waudby-Smith and Ramdas [37]. Anytime-valid comparison of forecasters was developed by Henzi and Ziegel [26] for binary outcomes and by Choe and Ramdas [9], whose confidence sequences require bounded score differences or a known global bound; they leave unbounded outcomes open and suggest the asymptotic confidence sequences of Waudby-Smith et al. [38], which we find lose error control at small α. Arnold et al. [2, Lemma B.1] bound quantile-score differences by a predictable multiple of the distance between the quantiles; Lemma 1 is the interval-score version, with the additional point that the deviation from a predictable centre vanishes off the miss event. What is new here is the sample-complexity analysis (lower bound, matching upper bound, the $1/\alpha$ separation between variance- and range-calibrated procedures, and the Diebold–Mariano mechanism) and the ES asymmetry.

2 The structure of tail score differences

Rounds are indexed by $t = 1, 2, \ldots$. Let $(\mathcal{G}_t)$ be a filtration in which $\mathcal{G}_t$ is the information available when the forecasts for round t are issued, and the outcome y_t is $\mathcal{G}_{t+1}$-measurable. Two forecasters A and B issue $\mathcal{G}_t$-measurable finite intervals $I_{A,t} = [l_{A,t}, u_{A,t}]$ and $I_{B,t} = [l_{B,t}, u_{B,t}]$. Multi-step forecasts fit this framework as long as the intervals for round t are issued before y_t is observed. Lower scores are better, and we write

$$d_t = \mathrm{IS}_\alpha(I_{A,t}; y_t) - \mathrm{IS}_\alpha(I_{B,t}; y_t), \qquad \Delta_t = \mathbb{E}[d_t \mid \mathcal{G}_t], \qquad \bar{\Delta}_t = \frac{1}{t}\sum_{s=1}^{t} \Delta_s.$$

The target $\bar{\Delta}_t$ is the average *conditional* predictive ability [9, 20]: $\bar{\Delta}_t > 0$ means that B has been better on average, round by round, given the information available just before each outcome. Let $H_t = \max\{|l_{A,t} - l_{B,t}|, |u_{A,t} - u_{B,t}|\}$ be the endpoint distance and $p_{A,t} = \mathbb{P}(y_t \notin I_{A,t} \mid \mathcal{G}_t)$, and similarly $p_{B,t}$, the conditional miscoverage probabilities. A *miss* is a round on which the outcome falls outside at least one of the two intervals, $y_t \notin I_{A,t} \cap I_{B,t}$; when one interval contains the other, the misses are those of the narrower one. For quantile and (VaR, ES) forecasts we call the corresponding rounds *exceedances*.

Lemma 1 (Structure of interval-score differences)**.** *Let $c_t = (u_{A,t} - l_{A,t}) - (u_{B,t} - l_{B,t})$ be the width difference, and*

$$m_t^- = \min\{0,\ l_{A,t} - l_{B,t},\ u_{B,t} - u_{A,t}\}, \qquad m_t^+ = \max\{0,\ l_{A,t} - l_{B,t},\ u_{B,t} - u_{A,t}\}.$$

Then, for every value of y_t,

1. *$d_t = c_t + (2/\alpha) f_t$, where f_t is the distance from y_t to $I_{A,t}$ minus its distance to $I_{B,t}$, and $m_t^- \le f_t \le m_t^+$, so that $a_t := c_t + (2/\alpha) m_t^- \le d_t \le b_t := c_t + (2/\alpha) m_t^+$;*
2. *$|f_t| \le H_t$, and $m_t^+ - m_t^- \le 2H_t$;*
3. *$f_t = 0$ if $y_t \in I_{A,t} \cap I_{B,t}$.*

The quantities c_t, a_t and b_t are $\mathcal{G}_t$-measurable. Consequently $\mathrm{Var}(d_t \mid \mathcal{G}_t) \le \mathbb{E}[(d_t - c_t)^2 \mid \mathcal{G}_t] \le (4H_t^2/\alpha^2) \min\{1, p_{A,t} + p_{B,t}\}$.

When the intervals are nested, $I_{A,t} \subseteq I_{B,t}$, one has $m_t^- = 0$, so $d_t \ge c_t$: the score difference can only rise above the width difference.

Quantile scores. For quantile forecasts $q_{A,t}, q_{B,t}$ at a lower-tail level $\tau < 1/2$, $\rho_\tau(y-q) = \tau(y-q) + (q-y)_+$ gives

$$d_t = \rho_\tau(y_t - q_{A,t}) - \rho_\tau(y_t - q_{B,t}) = \tau(q_{B,t} - q_{A,t}) + \{(q_{A,t} - y_t)_+ - (q_{B,t} - y_t)_+\}. \tag{2}$$

The braced term lies between $\min\{0, q_{A,t} - q_{B,t}\}$ and $\max\{0, q_{A,t} - q_{B,t}\}$ and vanishes unless $y_t < \max\{q_{A,t}, q_{B,t}\}$, an event of conditional probability at most $\max\{p_{A,t}, p_{B,t}\}$, where $p_{\cdot,t} = \mathbb{P}(y_t < q_{\cdot,t} \mid \mathcal{G}_t)$. Upper-tail levels are symmetric. Since $\mathrm{IS}_\alpha = (2/\alpha)\{\rho_{\alpha/2}(y-l) + \rho_{1-\alpha/2}(y-u)\}$, interval scores are sums of two such comparisons.

A common condition. Both cases are instances of the following structure, which is all the procedures below use.

Condition 1 (Predictable centre and range)**.** *There are $\mathcal{G}_t$-measurable $a_t \le c_t \le b_t$ with $a_t \le d_t \le b_t$. Write $D_t = d_t - c_t$, $R_t^- = c_t - a_t$ and $R_t^+ = b_t - c_t$.*

For the rates we use two further quantities: a bound $\bar{r}$ with $R_t^\pm \le \bar{r}$, and a bound π with $\mathbb{P}(D_t \neq 0 \mid \mathcal{G}_t) \le \pi$. Lemma 1 gives, for interval scores with $H_t \le H$ and $p_{A,t}, p_{B,t} \le C\alpha$,

$$\bar{r} = 2H/\alpha, \qquad \pi = 2C\alpha, \qquad \mathbb{E}[D_t^2 \mid \mathcal{G}_t] \le \bar{r}^2\pi = 8CH^2/\alpha, \tag{3}$$

and, for quantile scores at level τ with $|q_{A,t} - q_{B,t}| \le H$ and $p_{A,t}, p_{B,t} \le C\tau$, $\bar{r} = H$ and $\pi = C\tau$.

3 An anytime-valid confidence sequence

We now construct an empirical-Bernstein (EB) confidence sequence for $\bar{\Delta}_t$. The idea is to measure the score difference against the predictable centre c_t and to estimate its variance only from the deviations D_t, which are zero on every round without a miss. For a candidate value m of the average score difference and a tuning value $\lambda > 0$, the quantity $\exp\{\lambda \sum_s (d_s - m) - V\}$ is the wealth of a gambler who bets, with stake λ, that the score differences exceed m, and V is a penalty for their fluctuations; if the true average were m, this wealth could not grow large except with small probability. The lower bound L_t is the largest m that can be rejected in this way, using several stakes at once and a one-dimensional root search. Formally, let $g(u) = \{-\log(1-u) - u\}/u^2$ for $u \in (0,1)$, with $g(0) = 1/2$ and $g(u) = \infty$ for $u \ge 1$; g is increasing. Fix $\lambda_0 > 0$, set $\lambda_k = 2^{-k}\lambda_0$ for $k = 0, \dots, K-1$, and fix $\delta \in (0,1)$. For each k and $m \in \mathbb{R}$ let

$$V_{k,t} = \lambda_k^2 \sum_{s=1}^{t} g(\lambda_k R_s^-) D_s^2, \qquad \Phi_t(m) = \frac{1}{K} \sum_{k=0}^{K-1} \exp\Big\{\lambda_k \sum_{s=1}^{t} (d_s - m) - V_{k,t}\Big\},$$

where a component is set to zero from the first round s at which $\lambda_k R_s^- \ge 1$. The function Φ_t is continuous and decreasing in m. Define the lower bound L_t as the root of $\Phi_t(m) = 2/\delta$ (or $-\infty$ if every component is zero), and the upper bound U_t in the same way from $-d_t$, with the roles of R_t^- and R_t^+ exchanged.

Theorem 1 (Validity)**.** *Under Condition 1 alone,*

$$\mathbb{P}\big(\exists t \ge 1 : \ \bar{\Delta}_t \notin (L_t, U_t)\big) \le \delta.$$

The same holds for any countable grid (λ_k) with weights $w_k \ge 0$, $\sum_k w_k \le 1$, in place of the uniform weights $1/K$.

No moment, tail, coverage or endpoint-distance assumption enters: the ranges only need to be known before each outcome, and they are, because the intervals are. Each evaluation of Φ_t costs $O(K)$ once the running sums are updated, and the root is found by bisection. If λ_0, or the grid and weights, are computed from data observed up to some round t_0, the theorem applies conditionally on $\mathcal{G}_{t_0}$ to the rounds after t_0.

Choosing the grid. The uniform grid needs only the scale λ_0, which can be set from the ranges $R_t^{\mp}$ in a short warm-up segment before evaluation; we call the result the *untuned* sequence, and Theorem 2 applies to it. When the number of rounds n to be evaluated is known, the weights can instead concentrate on the λ that suits it, about $\{2\log(4/\delta)/(nv)\}^{1/2}$, where v, the mean of D_t^2, is estimated on an initial tuning segment. The resulting *tuned* sequence is valid for the rounds after that segment (Appendix B), which also lists the steps of its computation. When misses are rare, the ranges say little about v, and the uniform grid spends most of its weight on values of λ far from the best one; the tuned sequence avoids this at the cost of the tuning segment.

4 How many rounds a comparison needs

4.1 Upper bound

Theorem 2 (Detection time)**.** *Assume Condition 1, $R_t^{\pm} \le \bar{r}$ and $\mathbb{P}(D_t \ne 0 \mid \mathcal{G}_t) \le \pi$ for all t. Let $\delta \le 1/2$ and $t \ge 4$, and suppose $\lambda_0 \ge 1/(4\bar{r})$ and $2^{K-1} \ge \lambda_0 \bar{r}\sqrt{t}$. Let $\beta \in (0,1)$ and $\ell = \log(2K/\delta) + \log(2/\beta)$. If $\bar{\Delta}_t \ge \Delta > 0$ almost surely and*

$$t \ge \Big(96\,\frac{\bar{r}^2\pi}{\Delta^2} + 22\,\frac{\bar{r}}{\Delta}\Big)\ell, \tag{4}$$

then $\mathbb{P}(L_t > 0) \ge 1 - \beta$.

Substituting (3):

Corollary 1 (Interval scores)**.** *If $H_t \le H$ and $p_{A,t}, p_{B,t} \le C\alpha$ for all t, the confidence sequence separates the forecasters, $L_t > 0$, with probability at least $1-\beta$ as soon as*

$$t \ge \frac{\ell}{\alpha}\Big(768\,C\,\frac{H^2}{\Delta^2} + 44\,\frac{H}{\Delta}\Big).$$

At that round the expected number of misses, at most $2C\alpha t$, is of order $\ell C(CH^2/\Delta^2 + H/\Delta)$, which does not depend on α. For quantile scores at level τ with $|q_{A,t} - q_{B,t}| \le H$ and $p_{A,t}, p_{B,t} \le C\tau$, the condition is $t \ge \ell(96C\tau H^2/\Delta^2 + 22H/\Delta)$, with Δ in units of the quantile score.

For interval scores $|\Delta_t| \le 2H + (2/\alpha)H(p_{A,t} + p_{B,t}) \le (2+4C)H$, so $H/\Delta \le (2+4C)H^2/\Delta^2$, and the required number of rounds is of order $(1+C)H^2\ell/(\alpha\Delta^2)$. The ratio Δ/H is a dimensionless effect size: the average score difference relative to how far apart the forecasts are.

4.2 Lower bound

Theorem 3 (Lower bound for every test)**.** *Let $\tau \in (0, 1/2]$, $H > 0$ and $0 < \Delta \le \tau H$. There are fixed quantile forecasts $q_A < q_B = q_A + H$ at level $1-\tau$ and two distributions P_0, P_1 of an outcome with two support points such that, for i.i.d. outcomes:*

1. *the expected quantile-score difference is 0 under P_0 and Δ under P_1;*

2. *under both, each quantile is exceeded with probability at most* 2τ*;*

3. *every test based on* $y_1, \dots, y_T$ *whose errors under* P_0 *and* P_1 *are both at most* $\delta < 1/4$ *requires*

$$T \geq \tau(1-\tau)\frac{H^2}{\Delta^2}\log\frac{1}{4\delta}.$$

For interval scores at level $\alpha = 2\tau$*, taking* $I_A = [l, q_A]$ *and* $I_B = [l, q_B]$ *with a common lower endpoint below both support points gives a pair with endpoint distance* H*, miscoverage at most* α*, and expected interval-score differences* 0 *and* $\Delta_{\mathrm{IS}} = \Delta/\tau$*, and the bound reads* $T \geq 2(1 - \alpha/2)H^2 \log\{1/(4\delta)\}/(\alpha\Delta_{\mathrm{IS}}^2)$*.*

The lower bound applies to anytime-valid procedures as well, since the event that a level-δ confidence sequence excludes zero by round T is a test. It matches Corollary 1: both are of order $H^2\log(1/\delta)/(\alpha\Delta_{\mathrm{IS}}^2)$. In expected misses of the narrower interval the bound is $(1-\tau)(H/\Delta_{\mathrm{IS}})^2\log\{1/(4\delta)\}$: no procedure can separate two forecasters before observing a number of misses of order $(H/\Delta_{\mathrm{IS}})^2\log(1/\delta)$, however many rounds that takes. The outcome laws in the construction are bounded; rarity, not heavy tails, is the obstacle.

4.3 Range-calibrated procedures

A Hoeffding-type confidence sequence uses the same predictable ranges but not the variance: by Hoeffding's lemma, $\exp\{\lambda\sum_s(d_s - \Delta_s) - \lambda^2\sum_s(b_s - a_s)^2/8\}$ is a supermartingale for each $\lambda \geq 0$, and a mixture over λ gives bounds $(L_t^{\mathrm{H}}, U_t^{\mathrm{H}})$ in the same way as in Section 3.

Proposition 1. *If* $b_s - a_s = r$ *for all* s*, then for every grid and weights,* $L_t^{\mathrm{H}} \leq \bar{d}_t - r\sqrt{\log(2/\delta)/(2t)}$*, where* $\bar{d}_t$ *is the average of* $d_1, \dots, d_t$*. For nested intervals with* $H_t = H$*,* $r = 2H/\alpha$*, so the radius exceeds* 2Δ *while* $t < H^2\log(2/\delta)/(2\alpha^2\Delta^2)$*. If moreover the* d_t *are i.i.d. with mean* $\Delta > 0$ *and variance* σ^2*, then for such* t*,* $\mathbb{P}(L_t^{\mathrm{H}} > 0) \leq \sigma^2/(t\Delta^2)$*.*

In the Gaussian example below, σ^2/Δ^2 is of order $1/\alpha$, so the bound on $\mathbb{P}(L_t^{\mathrm{H}} > 0)$ is of order α at the threshold, and the Hoeffding-type sequence needs of order $1/\alpha$ times more rounds than the confidence sequence of Section 3.

The variance of d_t is of order H^2/α while its range is of order H/α; the gap between the two is the gap between the number of misses and the number of rounds.

4.4 A Gaussian example

Let $y \sim N(0,1)$, let z be its $(1-\alpha/2)$-quantile, and let B be the exact central interval $[-z, z]$. Forecaster A either narrows or widens it by η/z at each end, with $\eta = 1/2$. Then $H = \eta/z$, and Δ and $\mathrm{Var}(d)$ follow from one-dimensional integrals of the normal tail. As $\alpha \downarrow 0$, $\Delta \asymp 1/z$ and $\mathrm{Var}(d) \asymp 1/(\alpha z^2)$, so Δ/H tends to a positive constant and $\mathrm{Var}(d)/\Delta^2 \asymp 1/\alpha$; for the widened interval, for example, $\Delta \sim 2(\eta - 1 + e^{-\eta})/z$ and $\mathrm{Var}(d) \sim 8\{1 - (1+\eta)e^{-\eta}\}/(\alpha z^2)$. Section 7 shows both rates in this example.

5 The Diebold–Mariano test when misses are scarce

The Diebold–Mariano statistic is $\mathrm{DM}_T = \sqrt{T}\bar{d}_T/\hat{s}_T$, where $\hat{s}_T^2$ is a (heteroskedasticity and autocorrelation consistent) estimate of the long-run variance of d_t, and the test of equal predictive ability rejects at level γ when $|\mathrm{DM}_T| > z_{1-\gamma/2}$ [12, 32]. Its justification is a central limit theorem for $\bar{d}_T$, which needs many misses.

Proposition 2. *Let $(\sigma_t, \varepsilon_t)$ be i.i.d., with $\sigma_t > 0$ independent of ε_t, $\mathbb{E}\sigma_t^2 < \infty$, and $y_t = \sigma_t \varepsilon_t$, where σ_t is known when the forecasts are issued. Let $I_{A,t} = \sigma_t[-a, a]$ and $I_{B,t} = \sigma_t[-b, b]$ with $0 < a < b$, both scored by IS_α, and let $p = \mathbb{P}(|\varepsilon_t| > a)$. Let $\hat{s}_T^2$ be the sample variance or the Newey–West estimator with Bartlett weights, autocovariances with divisor T and lag $o(T)$, and declare A significantly better when $\mathrm{DM}_T < -z_{1-\gamma/2}$ (with $\mathrm{DM}_T = -\infty$ if $\bar{d}_T < 0 = \hat{s}_T$). Then*

$$\mathbb{P}(\mathrm{DM}_T < -z_{1-\gamma/2}) \geq (1-p)^T \, q_T,$$

where q_T depends only on the law of (σ_t), not on a, b or the law of ε_t; $q_T \to 1$ as $T \to \infty$, and $q_T = 1$ if σ_t is constant. In particular, if $a = a_T$ varies with T so that $Tp_T \to \mu$, where $p_T = \mathbb{P}(|\varepsilon_t| > a_T)$, the probability is at least $e^{-\mu} - o(1)$, for every value of $\bar{\Delta}_T$, including the null value 0 and values at which A is worse.

With few misses, the score difference is dominated by the width difference, which always favours the narrower interval, and the variance estimate reflects only the variation in widths, so the statistic is large. Where Theorem 3 says that no procedure can decide, the test decides, in a direction fixed by the widths. The bound comes from the event N_T that no round up to T is missed, and it identifies the mechanism rather than its size: in the Gaussian design of Section 7 with $\alpha = 0.01$ and $T = 250$, $(1-p)^T$ is below 0.1%, while the test favours the narrower interval in 12% of samples. With a handful of misses the variance estimate is still dominated by the width variation and by the few misses observed, and the distortion persists as long as the expected number of misses is small.

Proposition 3 (No valid test matches the Diebold–Mariano test)**.** *In the setting of Proposition 2 with $\sigma_t \equiv 1$, let P be any law of ε_t with $p = \mathbb{P}_P(|\varepsilon_t| > a) \geq \alpha$. Let ϕ be any test based on $y_1, \ldots, y_T$ that declares A better with probability at most δ under every i.i.d. law of ε_t under which A is not better. Then*

$$\mathbb{P}_P(\phi \text{ declares } A \text{ better}) \leq \delta + 1 - (1-p)^T \leq \delta + Tp,$$

whereas the Diebold–Mariano test declares A better with probability at least $(1-p)^T$, under P and under a law under which A is not better.

When the expected number of misses Tp is small, every valid test is nearly powerless, while the Diebold–Mariano test rejects with probability close to one whatever the truth: its additional rejections are errors, not power. The condition $p \geq \alpha$ only asks that the narrower interval not over-cover. The proposition applies to any procedure that controls its error exactly, including the confidence sequence of Section 3; procedures whose error control is only asymptotic, such as the corrections studied in Section 7, are not covered by it.

6 Expected shortfall

Expected shortfall (ES) is not elicitable on its own, but the pair of value-at-risk (VaR) and ES is [15], and (VaR, ES) forecasts are compared through a strictly consistent scoring function for the pair [33]. We work with returns y, whose lower tail is the loss tail, a level $\tau \in (0, 1/2)$, and forecasts (v, e) with $e \leq v < 0$ for the τ-quantile and the τ-expected shortfall, scored by the FZ0 loss of Patton et al. [34],

$$S(v, e; y) = -\frac{1}{\tau e}\mathbf{1}(y \leq v)(v - y) + \frac{v}{e} + \log(-e) - 1, \tag{5}$$

the member of the class of Fissler and Ziegel [15] whose differences are invariant to the scale of the returns. Write $\omega = 1/(\tau|e|)$ for its tail slope and $d_t = S(v_{A,t}, e_{A,t}; y_t) - S(v_{B,t}, e_{B,t}; y_t)$.

Lemma 2 (Structure of FZ0 differences)**.** *Let* $w(v,e) = v/e + \log(-e)$, $c_t = w(v_{A,t}, e_{A,t}) - w(v_{B,t}, e_{B,t})$ *and* $D_t = d_t - c_t$. *Then*

1. $D_t = \omega_{A,t}(v_{A,t} - y_t)_+ - \omega_{B,t}(v_{B,t} - y_t)_+$, *which is zero unless* $y_t \le \max\{v_{A,t}, v_{B,t}\}$;
2. *if* $e_{A,t} \ge e_{B,t}$, *so that A's ES is the less conservative, then* $D_t \ge -\omega_{B,t}(v_{B,t} - v_{A,t})_+$, *and if moreover* $e_{A,t} > e_{B,t}$, *then* $D_t \to \infty$ *as* $y_t \to -\infty$; *the case* $e_{A,t} \le e_{B,t}$ *is symmetric;*
3. w *is strictly increasing in* $|v|$ *and nondecreasing in* $|e|$, *so* $c_t \le 0$ *whenever A is less conservative in both components,* $v_{A,t} \ge v_{B,t}$ *and* $e_{A,t} \ge e_{B,t}$, *and* $c_t < 0$ *if moreover* $v_{A,t} > v_{B,t}$.

As for intervals and quantiles, the score difference departs from a value known in advance only on exceedance days (item 1). Unlike them, it is unbounded on one side whenever the ES forecasts differ (item 2). The same holds for every strictly consistent score in the characterization of Fissler and Ziegel [15]: the difference contains, for outcomes below both VaRs, a term proportional to $\{G_2(e_A) - G_2(e_B)\}\, y$ with G_2 strictly increasing.

What can be certified. On days on which A's ES is the less conservative, d_t is bounded below, so Theorem 1 applies to the lower bound; but large positive deviations inflate its variance term. Flooring the outcome repairs this without an assumption: replacing an extreme return by a less extreme one can only lower d_t on these days, so a lower bound computed from the floored differences remains a valid lower bound for the original ones. Let $\underline{y}_t \le \min\{v_{A,t}, v_{B,t}\}$ be $\mathcal{G}_t$-measurable, and let $\tilde{d}_t$ be the score difference evaluated at $\max\{y_t, \underline{y}_t\}$, which lies in a predictable bounded range.

Proposition 4 (One-sided certification)**.** *Let* $\mathcal{L} = \{t : e_{A,t} \ge e_{B,t}\}$, *a predictable set of days, and let* $\bar{\Delta}_n^{\mathcal{L}}$ *be the average of* $\Delta_t = \mathbb{E}[d_t \mid \mathcal{G}_t]$ *over the first n days in* $\mathcal{L}$. *The lower bound* L_n *of Section 3, computed from* $\tilde{d}_t$ *on the days in* $\mathcal{L}$, *satisfies* $\mathbb{P}(\exists n : \bar{\Delta}_n^{\mathcal{L}} < L_n) \le \delta/2$, *with no assumption on the distribution of the returns. The same holds for the upper bound on the days in the complement of* $\mathcal{L}$.

Thus one can show, without assumptions, that a forecaster whose ES is less conservative was worse on the days on which it was less conservative. The reverse cannot be shown without an assumption on the loss tail.

Proposition 5 (No test can certify the less conservative ES)**.** *Fix forecasts* (v_A, e_A) *and* (v_B, e_B) *with* $e_A > e_B$, *used on every day, and let the returns be i.i.d. Let* P_1 *be any law with a finite first moment under which* $\mathbb{E}_{P_1} d < 0$, *so that A is better. Let* ϕ *be any test based on* $y_1, \dots, y_T$ *that rejects the hypothesis* $\mathbb{E}d \ge 0$ *with probability at most* δ *under every law with a finite first moment for which that hypothesis holds. Then* $P_1(\phi \text{ rejects}) \le \delta$.

The proof mixes P_1 with a small probability ε of a very large loss, which reverses the comparison while changing the law of the sample by at most $T\varepsilon$ in total variation; this is the obstruction of Bahadur and Savage [3]. Since the perturbation changes the probability of an exceedance by at most ε, the result holds even if the hypothesis is restricted to laws whose exceedance probabilities for both VaR forecasts lie in any given open neighbourhood of those under P_1, for instance laws under which both VaR forecasts are nearly calibrated. Certifying that a less conservative ES forecast is better therefore requires an assumption on the loss tail. Under a known lower bound on the returns, flooring the outcome at that bound changes nothing, the score difference has a predictable range on all days, and the results of Sections 3 and 4 apply.

Rates and the Diebold–Mariano test. Both carry over. With equal ES forecasts $e_A = e_B = e$ and $v_B = v_A - H$, the FZ0 difference is $\omega\{-\tau H + \min((v_A - y)_+, H)\}$, the construction of Theorem 3 scaled by $\omega = 1/(\tau|e|)$, so separating the forecasters requires at least $(1-\tau)\{H/(|e|\Delta)\}^2 \log\{1/(4\delta)\}$ expected exceedances, where $H/|e|$ is the distance between the VaR forecasts relative to the ES. On a sample without exceedances the score difference equals c_t, which by item 3 of Lemma 2 favours the forecaster that is less conservative in both components, strictly when its VaR is strictly less conservative. The proof of Proposition 2 then applies with c_t in place of the width difference.

7 Simulations

Error control. In the first design the two forecasters have equal expected interval scores, so $\bar{\Delta}_t = 0$. Forecaster A issues $[-k_A\sigma_t, k_A\sigma_t]$, where k_A is 0.85 times the $(1-\alpha/2)$-quantile of the errors, and B issues the wider $[-k_B\sigma_t, k_B\sigma_t]$, with k_B chosen so that both expected scores coincide. Outcomes are standard Gaussian with $\sigma_t = 1$, or $y_t = \sigma_t\varepsilon_t$ with Student-t_3 errors and a lognormal scale σ_t (log-standard deviation 0.5) known to both forecasters. For each $\alpha \in \{0.2, 0.1, 0.05, 0.02, 0.01\}$ we run 1000 replications of length 4000 and evaluate the procedures at $T \in \{250, 1000, 4000\}$. Table 1 and Figure 1 report the results.

The empirical-Bernstein (EB) sequence never excluded zero in the Student-t design and did so in at most 0.1% of replications in the Gaussian design, and the Hoeffding sequence never did; the tuned sequence of Section 3, tuned on the first 10% of each horizon T and evaluated at T, erred in at most 0.3%. The asymptotic sequence of Waudby-Smith et al. [38] is set to be tightest at $t = 500$ and monitored from round 31, since its variance estimate is degenerate in the first rounds. It excluded zero at some round in 2.5–4.8% of Gaussian replications at $\alpha = 0.2$, but in 58–60% at $\alpha = 0.01$, and in 72–75% at $\alpha = 0.01$ in the Student-t design. Most of its failures occur early in the sequence, while few misses have occurred and the variance estimate is small; evaluated at the final round only, its error is at most 2.2% for $T \geq 1000$, but up to 11.6% at $T = 250$.

The two-sided Diebold–Mariano test at 5% has 2.5% nominal error in each direction. In favour of the narrower interval, its rejection rate at $T = 250$ is 12.0% (Gaussian) and 18.6% (Student-t) at $\alpha = 0.01$, and 7.3% and 13.5% at $\alpha = 0.02$. In favour of the wider interval it never exceeds 2.8%. Figure 1 plots these rates against the expected number of misses of the narrower interval: the points for different α fall on one curve, and the excess over the nominal level shrinks as that number grows. It exceeds twice the nominal level, 5%, in every case with fewer than 25 expected misses, and in the Student-t design also at 31, 35 and 63.

Small-sample corrections. We also apply three standard modifications of the test, each two-sided at 5%: the correction of Harvey et al. [25], which rescales the statistic and uses Student-t critical values; the fixed-smoothing test of Coroneo and Iacone [10], with a weighted periodogram variance estimate on the $\lfloor T^{1/3} \rfloor$ lowest Fourier frequencies and Student-t critical values; and a bootstrap-t test, which compares the statistic with the quantiles of its distribution over 499 resamples of the rounds, drawn independently as the outcomes are. Table 2 reports their rates in favour of the narrower interval. The first two change the rates by at most 1.5 points, since they correct the variance estimate, whereas the distortion comes from the skewness of d_t: most rounds favour the narrower interval by a small amount and a few favour the wider one by a large amount. The bootstrap-t test captures this skewness and keeps its rate at or below 3.6% in every case except the smallest, $T = 250$ in the Student-t design at $\alpha = 0.01$, with four expected misses, where it is 6.4%; in favour of the wider interval its rate is at most 3.7% (not shown). It carries no finite-sample guarantee, and Proposition 3 does not apply to it.

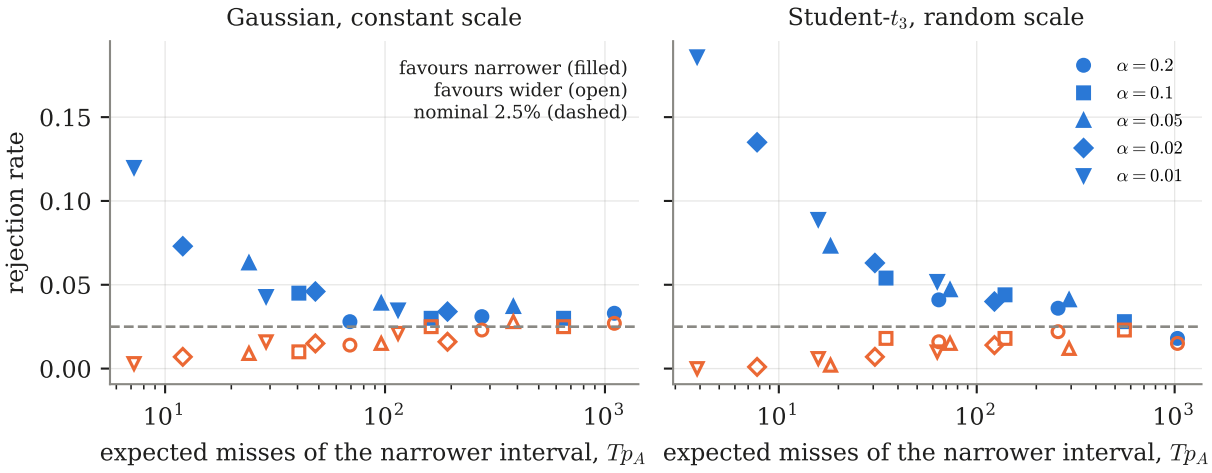


Figure 1: Rejection rates of the two-sided 5% Diebold–Mariano test under equal expected scores, against the expected number of misses of the narrower interval. Blue: the test declares the narrower forecaster better; orange: the wider. Marker shapes give α; the three points per shape are $T = 250$, 1000 and 4000.

Sample size. In the second design, outcomes are Gaussian, B is the exact central interval and A is narrowed or widened by η/z with $\eta = 1/2$ (Section 4.4), so that A is worse. For $\alpha \in \{0.2, 0.1, 0.05, 0.02, 0.01, 0.005\}$, 200 replications record the first round at which the EB lower bound exceeds zero. Because the ranges are constant, the Hoeffding radius is deterministic, and we report the round at which it falls below Δ. Detection times are recorded on a geometric grid of about 600 points.[1] Figure 2 also shows the sample size at which a fixed-sample one-sided test at level 2.5% with known variance has 80% power, $(z_{0.975} + z_{0.8})^2 \operatorname{Var}(d)/\Delta^2$, and the scale $H^2/(\alpha\Delta^2)$ of Theorem 3, where Δ is the expected interval-score difference (Δ_{IS} in that theorem).

The EB and fixed-sample curves are parallel to $H^2/(\alpha\Delta^2)$: the number of rounds grows like $1/\alpha$. The median number of misses at the EB detection time is nearly constant, between 191 and 237 for the narrowed interval and between 126 and 181 for the widened one, from $\alpha = 0.2$ to $\alpha = 0.005$. The Hoeffding sequence grows like $1/\alpha^2$: it needs 1.2–1.7 times as many rounds as EB at $\alpha = 0.2$ and 50–73 times as many at $\alpha = 0.005$, about 1.4 to 2.7 million rounds. Anytime validity costs EB a factor between 1.6 and 2.6 relative to a fixed-sample test that knows the variance and is valid only at the planned sample size.

With plentiful misses, as here, the Diebold–Mariano test is reliable: at the sample size T_{80} at which the known-variance test has 80% power, its power is 74–84% with no wrong-sign rejection, and that of the bootstrap-t test 68–88%, against 12–38% for the tuned sequence (Appendix B). Interpolating the power of the tuned sequence between T_{80}, $2T_{80}$ and $4T_{80}$ on a logarithmic scale, it reaches the power that the Diebold–Mariano test has at T_{80} after 1.9 to 3.3 times as many rounds. This factor is close to the ratio $\log(2/\delta)/(z^2_{0.975}/2) \approx 1.9$ of the exponents of a Chernoff bound and of the normal approximation, which is the price of a finite-sample guarantee.

[1]The reported detection time is the first grid round at which the lower bound exceeds zero, so it may exceed the exact first round by at most the spacing of the grid, about 2%.

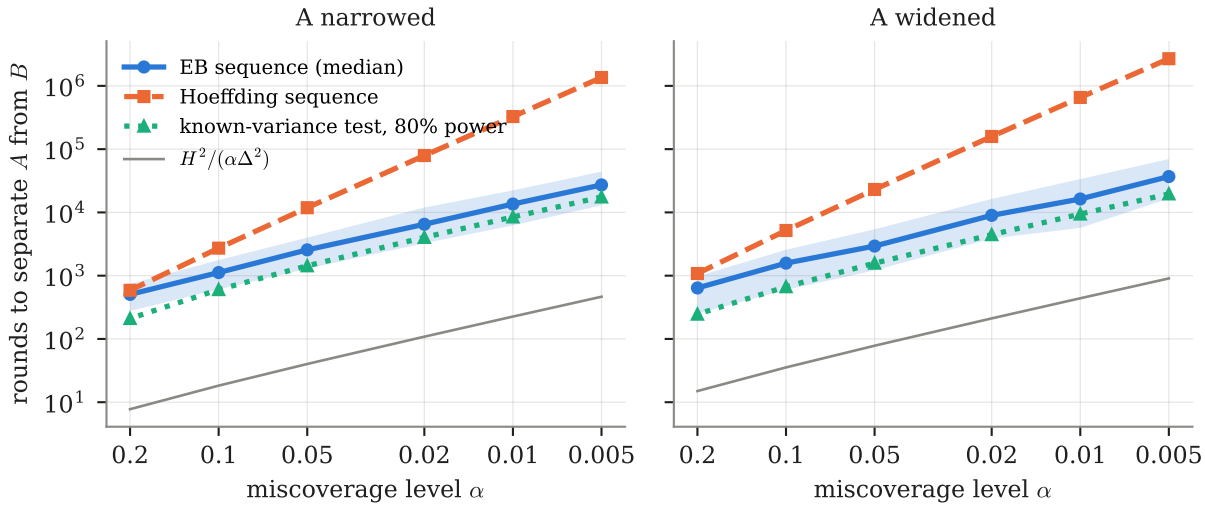


Figure 2: Rounds needed to separate a narrowed or widened Gaussian interval from the exact one. EB: median first round at which the 95% lower bound exceeds zero, with the 10%–90% band over 200 replications. Hoeffding: round at which its deterministic radius falls below Δ. Known-variance test: sample size for 80% power with known variance. Grey: the scale of the lower bound, without constants.

8 Application: interval forecasts at levels from 80% to 99%

Data. Online conformal methods provide many interval forecasters with the same coverage target, which differ mainly in how they track the tails; they are a natural test bed for comparisons at small α. We use seven public series: the oil temperature of an electricity transformer, hourly (ETTh1) and every 15 minutes (ETTm1) [42]; the occupancy rate of one road sensor, hourly, and one daily exchange rate [29]; German–Luxembourg day-ahead electricity prices, hourly, from Open Power System Data [39]; and the first twenty series of the M4 hourly [31] and NN5 daily [23] collections, each concatenated into one series. After 50 warm-up rounds, which initialise the methods and are not evaluated, the evaluation periods have between 5,538 and 66,146 rounds; they are shorter at the 99% level, for the reason given below.

Forecasters. All methods share one point forecast and one scale. The point forecast μ_t for y_t is the mean of five simple forecasters (naive, seasonal naive, rolling mean, simple exponential smoothing and a linear trend), each using data up to round $t-h$, where h is the seasonal period (24 for hourly series, 96 for ETTm1, 7 for daily series). The scale σ_t is the standard deviation of the last 50 observations, floored at 5% of the series' standard deviation. Each method turns the standardized scores $|y_s-\mu_s|/\sigma_s$, $s<t$, into an interval $[\mu_t-\sigma_t q_t^-, \mu_t+\sigma_t q_t^+]$, so the intervals for round t depend on information up to round $t-1$ only and satisfy the conditions of Section 2. The six methods are: adaptive conformal inference aggregated over step sizes, AgACI [41]; dynamically tuned adaptive conformal inference, DtACI [21]; strongly adaptive online conformal prediction, SAOCP [6]; quantile tracking (QT) and conformal PI control (PI) [1]; and AgACI-W, introduced here, which aggregates adaptive conformal experts over several reference windows as well as step sizes, with the ML-Poly rule of Gaillard et al. [19]. For the methods with a tuning parameter (the reference window of AgACI and DtACI, the learning rate of QT and PI) we keep the configuration

with the lowest mean score on the evaluation period. The comparisons are therefore between the best members of each family, selected in sample, and we use them to compare the procedures rather than to rank the methods. At the 99% level the candidate reference windows are longer, and because they need more history, the evaluation period of six series is 1,100 to 1,500 rounds shorter at that level.

Procedures. For each data set and level we compare all 15 pairs of methods. The Diebold–Mariano test is two-sided at 5% with a Newey–West variance and lag $\max\{h-1, \lfloor 4(T/100)^{2/9} \rfloor\}$. The confidence sequences use $\delta = 0.05$. The EB sequence is the tuned one of Section 3, tuned on the first 10% of the evaluation period. The Hoeffding sequence uses the uniform grid with λ_0 equal to the inverse of twice the median range over the 50 warm-up rounds that precede the evaluation period. We report whether each sequence excludes zero at the last round and, for every procedure, which method it favours. As a check that uses no model, we also split each evaluation period in halves, apply each procedure to the first half (the EB sequence tuned on the first 10% of that half), and record whether the sign of its significant results agrees with the mean score difference on the second half.

Results. Table 3 summarizes the 105 comparisons at each level. Misses, rounds on which at least one of the two intervals misses, make up about 1.3α of all rounds, so at the 99% level the median pair has 179 of them out of more than ten thousand. The share of pairs that each procedure separates falls with α, fastest for the range-calibrated Hoeffding sequence, which separates 44% of pairs at 80% and none at 99%, as Proposition 1 predicts. The EB sequence separates 70% at 80% and 18% at 99%, and the Diebold–Mariano test declares 82% and 54% significant; the two never disagree in direction. The untuned EB sequence, with λ_0 from the 50 warm-up rounds as for the Hoeffding sequence, separates 59% and 13%.

The conditions of Section 4 roughly hold. Realized miss rates have a median of 0.98α to 1.00α at every level (range 0.6α to 4.1α), and over windows of $20/\alpha$ rounds their 90th percentile has a median of about 1.1α and is at most 5.1α, so the conditional constant C is small. Endpoint distances, scaled by σ_t, have a 90th percentile about three times the median. The effect size Δ/H has a median of 0.35, 0.37, 0.36 and 0.50 at the four levels: it does not shrink with α, so the $1/\alpha$ growth of the required number of rounds applies to these data.

The split-sample check shows how reliable the significant results are. Of the Diebold–Mariano results that are significant on the first half, the share whose sign agrees with the second-half mean falls from 100% at 80% to 97%, 95% and 90% at 99%. The sign of all 149 significant first-half results of the EB sequence agrees, although at 99% it has 15 such results against 49 for the Diebold–Mariano test. Configurations were selected on the whole period, so the check is not fully out of sample.

9 Application: expected shortfall for seven stock indices

Data and forecasts. We use daily closing prices of seven equity indices (S&P 500, FTSE 100, DAX, Nikkei 225, Euro Stoxx 50, Hang Seng and Bovespa) from January 2000 to September 2026, and percentage log returns. Six standard models produce one-day-ahead (VaR, ES) forecasts at $\tau = 0.025$ and $\tau = 0.01$, each using returns up to the previous day: historical simulation on the last 250 and 500 returns (HS-250, HS-500); the exponentially weighted variance with decay factor 0.94 and normal quantiles (EWMA); GARCH(1,1) with normal (GARCH-N) and standardized Student-t innovations (GARCH-t); and filtered historical simulation (FHS), which multiplies the GARCH-N volatility by the empirical quantile and tail mean of the standardized residuals. The

GARCH models have zero mean and are re-estimated every 20 days on the last 1000 returns. After an estimation window of 1000 days and a further 250 days that fix the scale λ_0 of the untuned confidence sequence, the evaluation periods have between 3,638 and 5,541 days.

Procedures. For each index and level we compare the 15 pairs of models by the FZ0 loss. The Diebold–Mariano test is two-sided at 5% with a Newey–West variance and lag $\lfloor 4(T/100)^{2/9} \rfloor$. We apply it to the full evaluation period and to each non-overlapping window of 250 days, the length of the regulatory backtesting sample. In each window we record whether a significant result favours the model whose ES forecast was less conservative on average over the window, and whether its sign agrees with that of the mean loss difference over all other days of the evaluation period. The certification of Proposition 4 is applied on the days on which each model is the less conservative, with the outcome floored at $\underline{y}_t = \kappa \min\{e_{A,t}, e_{B,t}\}$ and $\kappa = 2$, that is, at twice the more conservative ES forecast; $\delta = 0.05$, $K = 20$, and on the full period the sequence is tuned on the first 10% of the selected days. A tuning segment of 10% of a 250-day window would contain less than one expected exceedance, so within windows we use the untuned sequence, with λ_0 from the 250 days that precede the evaluation period. The choice of κ affects power but not validity; we also report $\kappa = 1.5$ and $\kappa = 3$.

Results. Table 4 summarizes the results. A 250-day window contains a median of nine days on which the return falls below at least one of the two VaR forecasts at $\tau = 0.025$, and four at $\tau = 0.01$, against about six and 2.5 for one calibrated forecast, because two forecasts are involved and the models' exceedance rates, pooled over indices, are 1.0–1.4 times the nominal level at $\tau = 0.025$ and 1.1–2.1 times at $\tau = 0.01$. The Diebold–Mariano test is significant in 18% of the window–pair tests at both levels, and 73% (at $\tau = 0.025$) and 80% (at $\tau = 0.01$) of the significant results favour the less conservative model. Over the full period, with a median of 203 and 104 exceedances per pair, the corresponding shares are 67% and 51%, counting a model as less conservative when its ES was the less conservative on most days. The tilt towards the less conservative model is thus modest at $\tau = 0.025$, 73% in windows against 67% over the full period, and large at $\tau = 0.01$, 80% against 51%, where windows hold the fewest exceedances. The significant window results that favour the less conservative model contradict the ranking on the other days in 18.4% and 21.1% of cases; those that favour the more conservative model do so in 6.9% and 10.5%.

The window tests are dependent: the 15 pairs of an index share the returns, and windows of different indices overlap in time. To assess the sampling uncertainty of these shares, we assign each window to the calendar year of its midpoint and resample years with replacement, keeping every window of the drawn years (4000 resamples). The 95% percentile interval for the difference between the two wrong-sign rates is $[1.7, 19.7]$ percentage points at $\tau = 0.025$ and $[0.0, 18.9]$ at $\tau = 0.01$; for the share of significant results that favour the less conservative model it is $[54, 89]$% and $[67, 90]$%. The asymmetry is thus clear at $\tau = 0.025$ and borderline at $\tau = 0.01$, where significant results of either kind are fewer.

Replacing the normal critical values of the Diebold–Mariano test by those of a circular block bootstrap-t test (blocks of 10 days, 999 resamples, with the statistic and its replicates studentized by the variance of the block means) removes most of the tilt. The share of significant window tests is similar, 18% and 20%, but only 51% and 60% of them favour the less conservative model. Their wrong-sign rates are 12.9% and 14.2%, against 9.5% and 6.3% for those that favour the more conservative model, so at $\tau = 0.01$ results in favour of the less conservative model remain the less reliable.

The certification never succeeds within a window, as the lower bound of Section 6 leads one to

expect on four to nine exceedances. On the full period it certifies that the less conservative model was worse on its own days for 48% of pairs at $\tau = 0.025$ and 44% at $\tau = 0.01$. Across $\kappa \in \{1.5, 2, 3\}$ these shares range over 48–50% and 42–44%. The untuned sequence certifies 43% and 24%. When it certifies, the full-period Diebold–Mariano test is significant in the same direction in 48 of 50 and 36 of 46 pairs, and never in the opposite direction.

10 Discussion

In tail forecast comparisons the effective sample size is the number of exceedances. Three practical recommendations follow. First, report the number of rounds on which the score difference departs from its predictable value; it is computable from the forecasts and the outcomes. Second, do not use normal critical values for the Diebold–Mariano test when this number is small: in our simulations, its error in favour of the narrower interval exceeded twice the nominal level in every design with fewer than 25 expected misses of the narrower interval, and with heavy-tailed outcomes in some designs with up to 63. The corrections of Harvey et al. [25] and Coroneo and Iacone [10] did not change this. A bootstrap-t version, resampling blocks of rounds when the score differences are dependent, kept its error near the nominal level in all but the smallest design, with power similar to that of the test with normal critical values, slightly lower for the widened interval. In the ES data it removed most of the tilt towards the less conservative model, although at the 99% level its significant results in favour of that model were still the less reliable. It is the natural replacement when an asymptotic justification suffices. Third, when a guarantee that holds in finite samples, at all sample sizes and without distributional assumptions is needed, as when a comparison is monitored over time or its conclusion carries regulatory weight, use the confidence sequence of Section 3, tuned to the planned horizon when one is known; in our simulations the price was two to three times as many rounds as the Diebold–Mariano test needs for the same power.

For ES, 250-day windows hold too few exceedances for a ranking, and the Diebold–Mariano test with normal critical values fills the gap in favour of the less conservative model. Comparative backtests meant to reward a less conservative ES forecast should state the bound on the loss tail they assume.

The conditions of Theorem 2 are needed for the rate, not for validity. If endpoint distances are heavy-tailed, as when intervals are scaled by a heavy-tailed volatility, the rate depends on the realized ranges, and the Diebold–Mariano statistic may itself have a non-Gaussian limit [17]. The results extend to the weighted interval score of epidemic forecasting [7], which is a sum of interval scores and satisfies Condition 1 with the summed ranges.

Data and code availability

The data sets are public; Section 8 lists their sources, and the index prices of Section 9 are daily closes from Yahoo Finance, downloaded on 27 October 2026; the downloaded prices are included with the code. Code that reproduces every number, table and figure is provided as Supporting Information.

Conflict of interest

The author declares no conflict of interest.

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A Proofs

A.1 Proof of Lemma 1

The distance from y to $[l,u]$ is $O(y)=\max\{l-y,0,y-u\}$, and $\mathrm{IS}_\alpha=(u-l)+(2/\alpha)O$. For real numbers, $\min_i(x_i-z_i)\le\max_i x_i-\max_i z_i\le\max_i(x_i-z_i)$. Applying this to $f_t=O_A(y_t)-O_B(y_t)$, with the three terms paired in order, gives $m_t^-\le f_t\le m_t^+$. The distance to a set is 1-Lipschitz in the Hausdorff distance between sets, which for intervals equals H_t; hence $|f_t|\le H_t$, and $m_t^\pm$ are bounded in absolute value by H_t. Item 3 holds because both distances vanish. The variance bound follows from items 2 and 3 and the union bound.

A.2 Proof of Theorem 1

Fix $\lambda\ge 0$ and a round t, write $R=R_t^-$, $D=D_t$, and let $F_t=\exp\{\lambda(d_t-\Delta_t)-\lambda^2 g(\lambda R)D^2\}$ if $\lambda R<1$ and $F_t=0$ otherwise. We show $\mathbb{E}[F_t\mid\mathcal{G}_t]\le 1$. If $\lambda R\ge 1$ there is nothing to prove. If $0<\lambda R<1$, apply the inequality $\exp\{u\xi-\psi_E(u)\xi^2\}\le 1+u\xi$, where $\psi_E(u)=-\log(1-u)-u$, valid for $\xi\ge -1$ and $u\in[0,1)$ [14], with $\xi=D/R\ge -1$ and $u=\lambda R$:

$$\exp\{\lambda D-\lambda^2 g(\lambda R)D^2\}\le 1+\lambda D.$$

If $R=0$, then $D\ge 0$ and the same inequality holds with $g(0)=1/2$, because $e^{x-x^2/2}\le 1+x$ for $x\ge 0$. Multiplying by $\exp\{\lambda(c_t-\Delta_t)\}$ and taking conditional expectations, with $\mathbb{E}[D\mid\mathcal{G}_t]=\Delta_t-c_t$,

$$\mathbb{E}[F_t\mid\mathcal{G}_t]\le\{1+\lambda(\Delta_t-c_t)\}e^{-\lambda(\Delta_t-c_t)}\le 1.$$

Hence $M_t(\lambda)=\prod_{s\le t}F_s$ is a nonnegative supermartingale with $M_0=1$, and so is the mixture $M_t=\sum_k w_k M_t(\lambda_k)$. Ville's inequality [36] gives $\mathbb{P}(\exists t: M_t\ge 2/\delta)\le\delta/2$. Since $t\bar{\Delta}_t=\sum_{s\le t}\Delta_s$, $M_t=\Phi_t(\bar{\Delta}_t)$, and $\Phi_t(\bar{\Delta}_t)<2/\delta$ implies $\bar{\Delta}_t>L_t$. The upper bound is symmetric, and a union bound completes the proof.

A.3 Proof of Theorem 2

Write $\ell_1=\log(2K/\delta)$, $\ell_2=\log(2/\beta)$, so that $\ell=\ell_1+\ell_2$, and $Q_t=\sum_{s\le t}D_s^2$. Since $-R_s^-\le D_s\le R_s^+$, $|D_s|\le\bar{r}$.

Step 1: a deterministic bound on $\bar{d}_t-L_t$. If $\lambda_k\le 1/(2\bar{r})$, then $\lambda_k R_s^-\le 1/2$ for every s, so the kth component of Φ_t is never set to zero. Every term of Φ_t is nonnegative, so for each such k, $\Phi_t(m)\ge K^{-1}\exp\{\lambda_k t(\bar{d}_t-m)-V_{k,t}\}$, which is at least $2/\delta$ when $m\le\bar{d}_t-(\ell_1+V_{k,t})/(\lambda_k t)$. Hence

$$L_t\ge\bar{d}_t-\rho_t,\qquad \rho_t=\min_{k:\,\lambda_k\le 1/(2\bar{r})}\frac{\ell_1+V_{k,t}}{\lambda_k t}.$$

For these k, g being increasing, $V_{k,t}\le g(1/2)\lambda_k^2 Q_t$ with $g(1/2)=4(\log 2-1/2)<0.773$. Let $\lambda_*=\{\ell_1/(0.773\,Q_t)\}^{1/2}$ ($\lambda_*=\infty$ if $Q_t=0$), and let $\lambda_\sharp$ be the largest grid point not exceeding $1/(2\bar{r})$. Because $\lambda_0\ge 1/(4\bar{r})$ and the grid halves down to $\lambda_{K-1}\le 1/(\bar{r}\sqrt{t})\le 1/(2\bar{r})$ (for $t\ge 4$), $\lambda_\sharp\in[1/(4\bar{r}),1/(2\bar{r})]$.

If $\lambda_* \geq \lambda_\sharp$, take $\lambda_k = \lambda_\sharp$:

$$t\rho_t \leq \frac{\ell_1}{\lambda_\sharp} + 0.773\,\lambda_\sharp Q_t \leq 4\bar{r}\ell_1 + 0.773\,\lambda_* Q_t = 4\bar{r}\ell_1 + (0.773\,\ell_1 Q_t)^{1/2}.$$

If $\lambda_* < \lambda_\sharp$, note that $Q_t \leq t\bar{r}^2$ and $\ell_1 \geq \log 4 > 0.773$ (as $\delta \leq 1/2$) give $\lambda_* \geq 1/(\bar{r}\sqrt{t}) \geq \lambda_{K-1}$, so some grid point satisfies $\lambda_*/2 < \lambda_k \leq \lambda_*$, and

$$t\rho_t \leq \frac{2\ell_1}{\lambda_*} + 0.773\,\lambda_* Q_t = 3\,(0.773\,\ell_1 Q_t)^{1/2}.$$

In both cases

$$t\rho_t \leq 2.64\,(\ell_1 Q_t)^{1/2} + 4\bar{r}\ell_1. \tag{6}$$

Step 2: the size of Q_t. Let $Z_s = D_s^2/\bar{r}^2 \in [0,1]$. Convexity gives $e^z \leq 1 + (e-1)z$ on $[0,1]$, and $\mathbb{E}[Z_s \mid \mathcal{G}_s] \leq \mathbb{P}(D_s \neq 0 \mid \mathcal{G}_s) \leq \pi$, so

$$\mathbb{E}[e^{Z_s} \mid \mathcal{G}_s] \leq 1 + (e-1)\pi \leq e^{(e-1)\pi}.$$

Thus $\exp\{\sum_{s\leq t} Z_s - (e-1)\pi t\}$ has expectation at most one, and Markov's inequality gives, with probability at least $1-\beta/2$,

$$Q_t \leq \bar{r}^2\{(e-1)\pi t + \ell_2\}. \tag{7}$$

Step 3: the deviation of $\bar{d}_t$. The variables $\xi_s = \Delta_s - d_s = \mathbb{E}[D_s \mid \mathcal{G}_s] - D_s$ are martingale differences with $\xi_s \leq R_s^+ + R_s^- \leq 2\bar{r}$ and $\sum_{s\leq t}\mathbb{E}[\xi_s^2 \mid \mathcal{G}_s] \leq \sum_{s\leq t}\mathbb{E}[D_s^2 \mid \mathcal{G}_s] \leq t\bar{r}^2\pi =: v$. Freedman's inequality [18] gives $\mathbb{P}(\sum_{s\leq t}\xi_s \geq x) \leq \exp[-x^2/\{2(v+2\bar{r}x)\}]$. With $x = (2v\ell_2)^{1/2} + 4\bar{r}\ell_2$ the right-hand side is at most $\beta/2$, so with probability at least $1-\beta/2$,

$$\bar{d}_t \geq \bar{\Delta}_t - \Big(\frac{2\bar{r}^2\pi\ell_2}{t}\Big)^{1/2} - \frac{4\bar{r}\ell_2}{t}. \tag{8}$$

Step 4: combination. On the intersection of the events in (7) and (8), which has probability at least $1-\beta$, (6) and $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, $\sqrt{\ell_1\ell_2} \leq \ell/2$ and $\ell_1, \ell_2 \leq \ell$ give

$$L_t \geq \bar{\Delta}_t - \{\sqrt{2} + 2.64\sqrt{e-1}\}\Big(\frac{\bar{r}^2\pi\ell}{t}\Big)^{1/2} - (4 + 1.32 + 4)\frac{\bar{r}\ell}{t} \geq \Delta - 4.88\Big(\frac{\bar{r}^2\pi\ell}{t}\Big)^{1/2} - 9.32\,\frac{\bar{r}\ell}{t}.$$

Both subtracted terms are below $\Delta/2$ under (4), which proves $L_t > 0$.

A.4 Proof of Theorem 3

Write the score for level $1-\tau$ as $\rho_{1-\tau}(y-q) = \tau(q-y) + (y-q)_+$. Let $y = q_B$ with probability θ_j and $y = q_A - 1$ otherwise, under P_j. Then $d = \rho_{1-\tau}(y-q_A) - \rho_{1-\tau}(y-q_B) = -\tau H + H\mathbf{1}(y = q_B)$, so $\mathbb{E}_j d = H(\theta_j - \tau)$. Take $\theta_0 = \tau$ and $\theta_1 = \tau + \Delta/H \leq 2\tau$. Under P_j, q_A is exceeded with probability θ_j and q_B never. The Kullback–Leibler divergence between the two Bernoulli laws is bounded by their χ^2 divergence, $\mathrm{KL}(P_1\|P_0) \leq (\theta_1-\theta_0)^2/\{\theta_0(1-\theta_0)\} = \Delta^2/\{H^2\tau(1-\tau)\}$, and for product measures it is T times this. By the Bretagnolle–Huber inequality [35, Lemma 2.6], the sum of the two error probabilities of any test is at least $\frac{1}{2}\exp\{-T\,\mathrm{KL}(P_1\|P_0)\}$; if both are at most δ, then $T\,\mathrm{KL}(P_1\|P_0) \geq \log\{1/(4\delta)\}$. For the interval version, the lower endpoint is never crossed, so $\mathrm{IS}_\alpha(I_A; y) - \mathrm{IS}_\alpha(I_B; y) = (2/\alpha)d = d/\tau$.

A.5 Proof of Proposition 1

Each component of the mixture is $\exp\{\lambda t(\bar{d}_t - m) - \lambda^2 t r^2/8\}$ with $\lambda \geq 0$. If $m \geq \bar{d}_t$, it is at most one, and so is the mixture, which therefore stays below $2/\delta$. If $m < \bar{d}_t$, maximizing over λ bounds it by $\exp\{2t(\bar{d}_t - m)^2/r^2\}$, and so is the mixture, which reaches $2/\delta$ only if $\bar{d}_t - m \geq r\sqrt{\log(2/\delta)/(2t)}$. Hence $L_t^{\mathrm{H}} \leq \bar{d}_t - r\sqrt{\log(2/\delta)/(2t)}$. With $r = 2H/\alpha$ the radius is at least 2Δ when $t \leq H^2 \log(2/\delta)/(2\alpha^2\Delta^2)$, and then $L_t^{\mathrm{H}} > 0$ requires $\bar{d}_t - \Delta > \Delta$, which by Chebyshev's inequality has probability at most $\sigma^2/(t\Delta^2)$.

A.6 Proof of Proposition 2

On the event N_T that $|\varepsilon_t| \leq a$ for all $t \leq T$, which has probability $(1-p)^T$ and depends on (ε_t) only, every round falls inside both intervals and $d_t = c_t = -2(b-a)\sigma_t$. Then $\mathrm{DM}_T = -\sqrt{T}\bar{\sigma}_T/\hat{s}_{\sigma,T}$, where $\hat{s}_{\sigma,T}$ is the same variance estimator applied to (σ_t). By independence, conditioning on N_T does not change the law of (σ_t). Let $q_T = \mathbb{P}(\sqrt{T}\bar{\sigma}_T/\hat{s}_{\sigma,T} > z_{1-\gamma/2})$, which depends only on the law of (σ_t). The sample autocovariances satisfy $|\hat{\gamma}_k| \leq \hat{\gamma}_0$, so the Newey–West estimator with Bartlett weights and lag L_T satisfies $\hat{s}^2_{\sigma,T} \leq (2L_T + 1)\hat{\gamma}_0$, and $\hat{\gamma}_0 \leq T^{-1}\sum_{t\leq T}\sigma_t^2$ is bounded in probability because $\mathbb{E}\sigma_t^2 < \infty$. With $L_T = o(T)$, $\hat{s}^2_{\sigma,T} = o_p(T)$, while $\bar{\sigma}_T \to \mathbb{E}\sigma_1 > 0$ almost surely, so $\sqrt{T}\bar{\sigma}_T/\hat{s}_{\sigma,T} \to \infty$ in probability and $q_T \to 1$. If σ_t is constant, $\hat{s}_{\sigma,T} = 0$ and the test rejects by convention.

A.7 Proof of Proposition 3

Let Q agree with P on $[-a, a]$ and put the remaining mass p on $\pm(b+1)$, beyond both intervals. There $d = c + (2/\alpha)(b-a)$ with $c = -2(b-a)$, so $\mathbb{E}_Q d = 2(b-a)(p/\alpha - 1) \geq 0$, and A is not better under Q; hence $\mathbb{P}_Q(\phi \text{ declares } A \text{ better}) \leq \delta$. Draw ε_t from P and replace each value outside $[-a, a]$ by a draw from the tail of Q; this couples P^T and Q^T so that the samples coincide on the event N_T of no miss, and the total variation distance between P^T and Q^T is at most $1 - (1-p)^T \leq Tp$. The bound for ϕ follows. On N_T, which has probability $(1-p)^T$ under both laws, the Diebold–Mariano test declares A better by the proof of Proposition 2.

A.8 Proofs for Section 6

Lemma 2. Item 1 is (5). For item 2, D_t is piecewise linear in y_t, equal to zero above both VaRs; below both it has slope $\omega_{B,t} - \omega_{A,t} \leq 0$ in y_t, so it increases as y_t decreases, and its minimum is attained at $y_t = \min\{v_{A,t}, v_{B,t}\}$ or above, where it equals $-\omega_{B,t}(v_{B,t} - v_{A,t})_+$ or zero. For item 3, $\partial w/\partial v = 1/e < 0$, and $\partial w/\partial e = (e-v)/e^2 \leq 0$ for $e \leq v$. Moving from (v_A, e_A) to (v_A, e_B) keeps $e \leq e_A \leq v_A$ and does not decrease w; moving on to (v_B, e_B) increases w, strictly if $v_A > v_B$.

Proposition 4. On a day in $\mathcal{L}$, D_t is nonincreasing in y_t below $\min\{v_{A,t}, v_{B,t}\}$ by Lemma 2, so $\tilde{d}_t \leq d_t$ and $\mathbb{E}[\tilde{d}_t \mid \mathcal{G}_t] \leq \Delta_t$. The process of the proof of Theorem 1, built from $\tilde{d}_t$ and with factor one on days outside $\mathcal{L}$, is a nonnegative supermartingale because $\mathcal{L}$ is predictable, and the argument of that theorem bounds the average of $\mathbb{E}[\tilde{d}_t \mid \mathcal{G}_t]$, hence of Δ_t, from below. The statement for the complement is symmetric.

Proposition 5. For $\varepsilon \in (0,1)$ and $M > 0$ let $P_{\varepsilon,M}$ be the mixture of P_1, with weight $1-\varepsilon$, and a point mass at $-M$, with weight ε. By Lemma 2, d evaluated at $y = -M$ grows linearly in M with slope $\omega_A - \omega_B > 0$, so for M large enough $\mathbb{E}_{P_{\varepsilon,M}} d \geq 0$ and $P_{\varepsilon,M}$ belongs to the hypothesis. The total variation distance between the T-fold products of P_1 and $P_{\varepsilon,M}$ is at most $T\varepsilon$. Hence $P_1(\phi \text{ rejects}) \leq \delta + T\varepsilon$ for every $\varepsilon > 0$.

B The tuned sequence and its power

The tuned sequence estimates v, the mean of D_t^2, and the 90th percentiles $r^{\mp}$ of $R_t^{\mp}$ on the tuning segment. For each side it sets $\lambda_* = \{2\log(4/\delta)/(nv)\}^{1/2}$, capped at $1/(2r^{\mp})$, with n the number of remaining rounds, and uses the grid $\lambda_* 2^j$, $j = -10, \ldots, 9$, with weight $1/2$ on $j = 0$ and $1/38$ on each other point. For a component with weight w, the lower bound is at least $\bar{d}_t - \{\log(2/(\delta w)) + V_{k,t}\}/(\lambda_k t)$, which is smallest near this λ_* when $V_{k,t} \approx \lambda_k^2 nv/2$. Validity follows from Theorem 1, since the weights are fixed before the remaining rounds are observed; the target is the average of Δ_t over those rounds.

In practice:

1. compute d_t, c_t, a_t and b_t for every round from the forecasts and the outcomes (Lemma 1 for intervals, (2) for quantiles, Lemma 2 for (VaR, ES));
2. on the tuning segment, set v to the mean of D_t^2 and $r^{\mp}$ to the 90th percentiles of $R_t^{\mp}$, and form the grid and weights above for each side;
3. on the remaining rounds, maintain $\sum_s d_s$ and, for each k, $\sum_s g(\lambda_k R_s^{\mp}) D_s^2$, and find L_t and U_t by bisection on m in $\Phi_t(m) = 2/\delta$;
4. declare B better once $L_t > 0$, and A better once $U_t < 0$.

The code accompanying the paper implements these steps.

Table 5 compares its power with that of the Diebold–Mariano test and its bootstrap-t version in the Gaussian design of Section 7.

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Table 1: Error rates (%) under equal expected scores. DM: two-sided 5% Diebold–Mariano test at T, split by the forecaster it declares better. Error by T: the 95% confidence sequence excludes zero at some round up to T (asymptotic: from round 31). Tp_A: mean number of misses of the narrower interval. 1000 replications.

			DM at T		error by T			
design	α	T	narrower	wider	EB	Hoeffding	asympt.	Tp_A
Gaussian	0.2	250	2.8	1.4	0.0	0.0	2.5	69
		1000	3.1	2.3	0.0	0.0	4.0	276
		4000	3.3	2.7	0.0	0.0	4.8	1104
	0.1	250	4.5	1.0	0.0	0.0	8.9	41
		1000	3.0	2.5	0.1	0.0	10.4	162
		4000	3.0	2.5	0.1	0.0	11.5	649
	0.05	250	6.3	0.9	0.0	0.0	20.5	24
		1000	3.9	1.5	0.0	0.0	21.8	96
		4000	3.7	2.8	0.0	0.0	23.4	383
	0.02	250	7.3	0.7	0.0	0.0	41.2	12
		1000	4.6	1.5	0.1	0.0	42.6	48
		4000	3.4	1.6	0.1	0.0	43.6	192
	0.01	250	12.0	0.3	0.0	0.0	57.5	7
		1000	4.3	1.6	0.0	0.0	59.2	29
		4000	3.5	2.1	0.0	0.0	59.8	115
t_3	0.2	250	4.1	1.6	0.0	0.0	4.5	64
		1000	3.6	2.2	0.0	0.0	6.6	258
		4000	1.8	1.5	0.0	0.0	7.1	1033
	0.1	250	5.4	1.8	0.0	0.0	12.5	35
		1000	4.4	1.8	0.0	0.0	13.8	139
		4000	2.8	2.3	0.0	0.0	15.3	557
	0.05	250	7.3	0.2	0.0	0.0	30.3	18
		1000	4.7	1.5	0.0	0.0	32.9	73
		4000	4.1	1.2	0.0	0.0	33.6	293
	0.02	250	13.5	0.1	0.0	0.0	51.6	8
		1000	6.3	0.7	0.0	0.0	54.3	31
		4000	4.0	1.4	0.0	0.0	55.0	123
	0.01	250	18.6	0.0	0.0	0.0	72.2	4
		1000	8.9	0.6	0.0	0.0	74.5	16
		4000	5.2	1.0	0.0	0.0	75.1	63

Table 2: Rejection rates (%) in favour of the narrower interval under equal expected scores, in the design of Table 1. DM: Diebold–Mariano test with normal critical values; HLN: with the correction of Harvey et al. [25]; WPE: fixed-smoothing test of Coroneo and Iacone [10]; bootstrap: bootstrap-t test. All two-sided at 5%, so the nominal rate is 2.5%. 1000 replications.

design	α	T	Tp_A	DM	HLN	WPE	bootstrap
Gaussian	0.2	250	69	2.8	2.8	2.5	1.9
		1000	276	3.1	3.1	3.1	3.0
		4000	1104	3.3	3.3	3.3	3.6
	0.1	250	41	4.5	4.4	4.2	1.9
		1000	162	3.0	3.0	2.5	2.4
		4000	649	3.0	3.0	2.8	2.7
	0.05	250	24	6.3	6.1	4.9	0.9
		1000	96	3.9	3.9	3.9	2.5
		4000	383	3.7	3.7	3.4	2.9
	0.02	250	12	7.3	7.3	7.4	1.0
		1000	48	4.6	4.5	4.4	1.7
		4000	192	3.4	3.4	3.2	2.8
	0.01	250	7	12.0	11.9	10.8	0.8
		1000	29	4.3	4.3	4.0	1.2
		4000	115	3.5	3.5	2.9	1.5
t_3	0.2	250	64	4.1	4.1	3.3	2.6
		1000	258	3.6	3.6	3.9	3.0
		4000	1033	1.8	1.8	2.3	2.4
	0.1	250	35	5.4	5.4	4.7	2.1
		1000	139	4.4	4.4	4.4	3.2
		4000	557	2.8	2.8	2.4	2.6
	0.05	250	18	7.3	7.3	6.6	2.3
		1000	73	4.7	4.7	4.4	2.6
		4000	293	4.1	4.1	3.9	3.1
	0.02	250	8	13.5	13.3	12.0	3.0
		1000	31	6.3	6.3	5.8	2.9
		4000	123	4.0	4.0	3.8	2.5
	0.01	250	4	18.6	18.4	17.4	6.4
		1000	16	8.9	8.9	8.3	1.1
		4000	63	5.2	5.2	4.8	2.9

Table 3: Comparisons of six online conformal methods on seven data sets, 15 pairs each. Misses: rounds on which at least one interval misses, median per pair. Significant: the procedure separates the pair over the evaluation period (DM: two-sided 5%; EB: tuned 95% confidence sequence; Hoeffding: 95% Hoeffding-type sequence). First half: number of significant results on the first half of the evaluation period, and the share of those whose sign agrees with the mean on the second half.

		significant (%)			first half			
					DM		EB	
level	misses	DM	EB	Hoeffding	signif.	sign agrees (%)	signif.	sign agrees (%)
80%	3905	81.9	69.5	43.8	72	100.0	58	100.0
90%	1982	66.7	47.6	26.7	66	97.0	40	100.0
95%	978	64.8	39.0	17.1	59	94.9	36	100.0
99%	179	54.3	18.1	0.0	49	89.8	15	100.0

Table 4: Comparisons of six (VaR, ES) models on seven equity indices, 15 pairs each, by the FZ0 loss. Windows: non-overlapping 250-day windows. Exceedances: days on which the return falls below the larger VaR forecast. DM: two-sided 5% Diebold–Mariano test with normal critical values. Wrong sign: share of the significant window results, of each kind, whose sign is opposite to that of the mean over all other days. EB certifies: share for which the one-sided 95% bound of Proposition 4 certifies that the less conservative model was worse on its own days, with the outcome floored at twice the more conservative ES; untuned sequence in windows, tuned sequence over the full period.

	$\tau = 0.025$	$\tau = 0.01$
250-day windows		
window–pair tests	2130	2130
median exceedances	9	4
DM significant (%)	17.8	18.0
favouring the less conservative (%)	73.1	80.2
wrong sign, favouring less conservative (%)	18.4	21.1
wrong sign, favouring more conservative (%)	6.9	10.5
EB certifies (%)	0.0	0.0
full period		
pairs	105	105
median exceedances	203	104
DM significant (%)	77.1	71.4
favouring the less conservative (%)	66.7	50.7
EB certifies (%)	47.6	43.8

Table 5: Power (%) to declare the worse forecaster A worse in the Gaussian design of Figure 2, at horizons T_{80}, $2T_{80}$ and (for EB) $4T_{80}$, where T_{80} gives the known-variance test 80% power. DM: two-sided 5% Diebold–Mariano test; boot: its bootstrap-t version; EB: the tuned sequence, tuned on the first 10% of the horizon. Neither DM nor boot declared A better in any replication. 200 replications.

	A narrowed							A widened						
	$T = T_{80}$			$T = 2T_{80}$			$4T_{80}$	$T = T_{80}$			$T = 2T_{80}$			$4T_{80}$
α	DM	boot	EB	DM	boot	EB	EB	DM	boot	EB	DM	boot	EB	EB
0.2	79.0	83.0	12.5	98.5	99.0	56.0	98.0	79.0	70.5	16.0	97.5	96.0	48.5	91.5
0.1	79.0	83.5	24.0	98.5	99.0	76.0	99.5	74.0	68.5	18.5	95.5	94.5	46.5	94.5
0.05	81.5	83.0	28.5	99.0	99.5	75.5	99.0	83.0	76.5	17.5	96.0	95.5	64.0	98.5
0.02	79.5	82.5	34.5	97.0	98.5	77.5	100.0	80.0	75.0	20.5	97.0	95.0	63.0	98.5
0.01	81.0	84.5	37.5	99.5	100.0	82.5	100.0	80.0	73.0	28.5	96.5	94.5	58.5	98.0
0.005	83.5	87.5	33.0	99.0	99.5	88.0	99.0	75.5	69.5	28.0	95.0	95.0	66.5	96.5